AUTOMOTIVE-GRADE (→) UFS AND E.MMC EMBEDDED FLASH DRIVES



SANDISK™

"BROCHURE"

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SANDISK™

The Storage You Can Drive On

Designed for the connected and autonomous car market, Sandisk's automotive-grade iNAND® UFS and e.MMC Embedded Flash Drives (EFDs) deliver high-performance and high-reliability storage for a variety of demanding automotive applications. The devices address the evolving needs of traditional applications such as infotainment and vehicle navigation systems, and next generation applications such as Vehicle-to-Vehicle/Infrastructure communications, telematics gateways, digital cluster, drive recorders, Advance Driver Assistance Systems (ADAS) and autonomous driving.

Smart partitioning, a sophisticated read refresh algorithm, power fail immunity, and an LDPC ECC engine are just a few of the features that make these some of the most advanced EFDs on the market. With additional industry-leading features including automotive-optimized 3D technology, iNAND® SmartSLC™ technology, and deep diagnostics, SANDISK® iNAND® automotive-grade e.MMC and UFS EFDs empower your data-driven applications.



3D Technology

- Higher performance, lower cost and lower power consumption
- Larger effective cell size and less cell-to-cell interference than 2D NAND
- Higher reliability margins when compared to 2D NAND with similar bit density

SmartSLC™ & Write Booster[1]

Sandisk's automotive grade iNAND® products feature industry-leading SmartSLC™ technology for e.MMC and UFS2.1 and Write Booster for UFS3.1 and UFS4.1, designed to boost host write performance and leverage the benefits of SLC to maximize device throughput and endurance.

Adaptive mechanisms enable writes to be routed through SLC blocks for increased write speeds and efficient migration of data to TLC to help ensure performance consistency.

Advantages:

Performance

- Boost sequential and random write performance to SLC level
- SLC-grade latency for better write-driven use cases
- Maintain high performance when media is fragmented
- Persistent performance even when media is full

Endurance

- Reduce host writes to the TLC area
- Reduces Write Amplification Factor (WAF) to the TLC area
- Increase product endurance and longevity (TBW)

Robustness

- Leverage SLC's higher resiliency to data corruption
- TLC/SLC redundancy increases power failure handling

Advanced Device Report

iNAND® branded products offer a proprietary device report feature with detailed information and diagnostics on the firmware and device status at runtime. This information provides a deeper EFD status understanding, on top of the e.MMC and UFS JEDEC specs in an easy-to-use method including:

- Wear-leveling areas and device health
- Voltage droop and occurrences
- Power diagnostics
- WAF (Write Amplification Factor) value
- Lifetime prediction
- Firmware release and update
- Host read and write
- Error detection and correction
- Temperature
- Failures and recovery

Smart Partitioning

iNAND® automotive-grade e.MMC and UFS Smart Partitioning implementation creates dedicated, physically separated and individually managed memory areas. This protects specific partitions from the unwanted effects of overuse as well as preventing any impact to the data integrity caused by activities in other areas, thus helping to maximize endurance, data retention and separation.

- TLC/MLC based User Area
- SLC based High Data Retention / Read Intensive / Secured area
- SLC based High Endurance area, tailored for write-intensive applications

Auto/Manual Read Refresh

A sophisticated automatic read refresh algorithm is implemented in the iNAND® automotive-grade e.MMC and UFS devices to offer better handling of data retention, read disturb and read refresh. It senses early signs of block degradation and automatically refreshes the data. The algorithm includes multiple types of scans designed to identify affected pages with high Bit Error Rate (BER).

In addition, the host can initiate Refresh via command, which is proprietary in iNAND® automotive-grade e.MMC and according to JEDEC UFS3.0 and JEDEC UFS4.1 in iNAND® automotive-grade UFS device.

Power Immunity

iNAND® automotive-grade e.MMC and UFS devices implement advanced mechanisms for protecting the device and the user data from data loss, data corruption or device damage due to external power failures. This addresses the occurrence of both write abort and voltage droop.

100% Content Preloading and Integrity Post IR Reflow

iNAND® automotive-grade e.MMC and UFS devices are designed to support 100% content preloading (via external programmer) pre-IR reflow and guarantee high data integrity post-IR reflow. This helps assure that the high temperature experienced during the IR-reflow process does not affect the data integrity of the preloaded data.

Thermal Management

To prevent the damage to the NAND or controller while operating in high temperatures, the iNAND® automotive-grade e.MMC and UFS devices implement a robust thermal management mechanism. This mechanism is activated by default and automatically adjusts the internal power mode to help protect from overheating. In addition, the iNAND® automotive-grade UFS device supports Temperature Event Notification- a JEDEC UFS3.0 and UFS4.1 parameter.

Suited for Automotive Applications

iNAND® automotive-grade e.MMC and UFS devices undergo rigorous qualification and production testing, and comply with automotive standards such as AEC-Q100 and are qualified to meet and exceed the standard requirements.

Fast Boot

iNAND® automotive-grade e.MMC and UFS devices enable the host to boot from boot partitions/LUs of the device.

This feature allows the system to be functional at the proper time. For example, the console display and the remaining car system functionality are needed the moment the driver enters or activates the vehicle. In addition, this feature enables the replacement of SPI NOR for the inherent capability of the iNAND® device, reducing BOM costs and board space.

Automotive-Grade Embedded Flash Drives

	 **iNAND® AT EU752** Grade 2	 **iNAND® AT EU552** Grade 2/3	 **iNAND® AT EU312** Grade 2/3	 **iNAND® AT EM132** Grade 2/3	 **iNAND® AT EM122** Grade 2/3
Capacity[2]	256GB – 1TB	64GB - 512GB	16GB - 256GB	32GB - 256GB	8GB - 64GB
Interface	UFS 4.1	UFS 3.1	UFS 2.1	e.MMC 5.1 HS400	e.MMC 5.1 HS400
Operating Temperature	ZA: -40°C to 105°C	ZA: -40°C to 105°C XA: -40°C to 85°C	ZA: -40°C to 105°C XA: -40°C to 85°C	ZA: -40°C to 105°C XA: -40°C to 85°C	ZA: -40°C to 105°C XA: -40°C to 85°C
Sequential R/W (MB/s)[3]	Up to 4200/3900	Up to 1600/1200	Up to 800/550	Up to 310/250	Up to 300/125
Random R/W (IOPS)[3]	Up to 420K/440K	Up to 100K/95K	Up to 45K/40K	Up to 20K/12K	Up to 22K/12K
Enhanced Features	Advanced Health Report, Smart Partitioning, Auto and Manual Refresh, Thermal Management, UFS 4.1 automotive features included	Advanced Health Report, Smart Partitioning, Auto and Manual Refresh, Thermal Management, UFS 3.0 automotive features included	Advanced Health Report, Smart Partitioning, Auto and Manual Refresh, Thermal Management, UFS 3.0 automotive features included	Advanced Health Report, Smart Partitioning, Auto and Manual Refresh, Thermal Management	Advanced Health Report, Smart Partitioning, Auto and Manual Refresh, Thermal Management
Package	11.5 × 13 × 1.2mm	11.5 × 13× 1.2mm	11.5 x 13 × 1.2mm	11.5 × 13 × 1.0mm, 11.5 × 13 × 1.2mm	11.5 × 13 × 0.8mm, 11.5 × 13 × 1.0mm, 11.5 × 13 × 1.2mm
Ordering Information	SDINHDI6-####-ZA	SDINFDQ6-##G-ZA1 SDINFDQ6-##G-XA1	SDINDDH6-##G-ZA2 SDINDDH6-##G-XA2	SDINBDA6-##G-ZA1 SDINBDA6-##G-XA1	SDINBDG4-##G-ZA3 SDINBDG4-##G-XA3

1. SmartSLC applicable for iNAND® AT EM132 and AT EU312 devices, Write Booster applicable for iNAND(r) AT EU552 and AT EU752 devices.
2. 1GB = 1 billion bytes, 1TB = 1 trillion bytes and 1PB= 1 quadrillion bytes. Actual user capacity may be less depending on operating environment.
3. Speeds vary by product and capacity. 1 MB/s = 1 million bytes per second. Based on internal testing; performance may vary depending upon host device, usage conditions, drive capacity, and other factors.



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